# MindSpace

A Mental-Health Support Chatbot

Data-Science Project Notes

An intent classifier (TF-IDF + neural network) that routes each message to either an empathetic reply learned from real counseling conversations, or an evidence-based answer retrieved from official WHO mental-health guidelines.

*Plain-English explanation of what the system does, how every part works, and how to defend it in a viva.*

# 1. What is MindSpace (in one minute)

MindSpace is a web chatbot that listens to how a person is feeling and replies supportively. You type (or speak) a message; the app figures out the **intent** behind it — for example anxiety, low mood, stress, or a general question — and how **confident** it is about that guess. Based on that confidence it picks one of two ways to answer.

It is built as a **data-science project**, not a black box. Every stage uses real, public data and standard, explainable techniques: we clean a labelled emotion dataset, turn text into numbers with TF-IDF, train a small neural network to recognise intent, measure how good it is with proper train/test metrics, and connect it to a searchable library of official World Health Organization (WHO) guidance.

## The two answer paths

- **Confident** (the model is sure what the user means): reply with an **empathetic message** drawn from real counseling conversations, after tone-correction and anti-repetition checks.
- **Unsure** (the model is not sure): **search the WHO knowledge base** for the most relevant passage and show that — but only if it clears a relevance bar; otherwise fall back to gentle support.

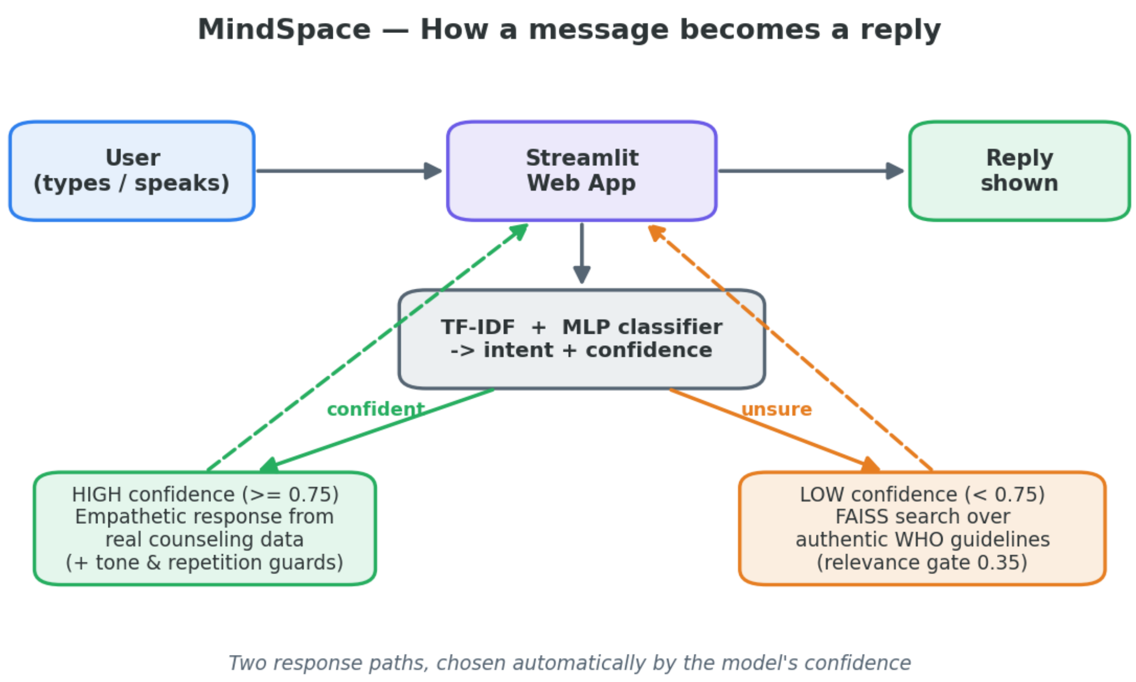


*Figure 1 — From a typed message to a reply. The classifier's confidence decides which of the two paths is used.*

## 2. How this matches the project proposal

The proposal asked for a complete data-science pipeline — real data, exploratory analysis and cleaning, NLP preprocessing, a trained model with a proper train/test evaluation, and visual results — applied to a useful problem. MindSpace delivers exactly that, and adds a retrieval layer so answers can be grounded in authoritative sources.

| Proposal requirement | How MindSpace meets it |
|---|---|
| Real dataset | dair-ai/emotion (~16k labelled messages) for training; Amod mental-health counseling conversations for replies; 4 official WHO PDFs for the knowledge base. |
| EDA & cleaning | Null/duplicate removal, very-short-row filtering, class balance inspection, text-length checks. |
| NLP preprocessing | Tokenisation, stop-word removal, lemmatisation (NLTK). |
| Feature engineering | TF-IDF with 1–2 word grams, 5,000 features. |
| Model + train/test | MLP neural-net classifier, stratified 80/20 split, class weights for imbalance. |
| Evaluation & visuals | Accuracy, precision/recall/F1 per class, confusion matrix, confidence histogram with the 0.75 threshold. |
| Real-world value | Confidence-routing to authentic WHO guidance keeps answers safe and grounded. |

# 3. The machine-learning pipeline (built offline, once)

This is the part that *learns*. It runs once on our machine, produces two saved files, and then the app just loads them — no training happens while a user chats.

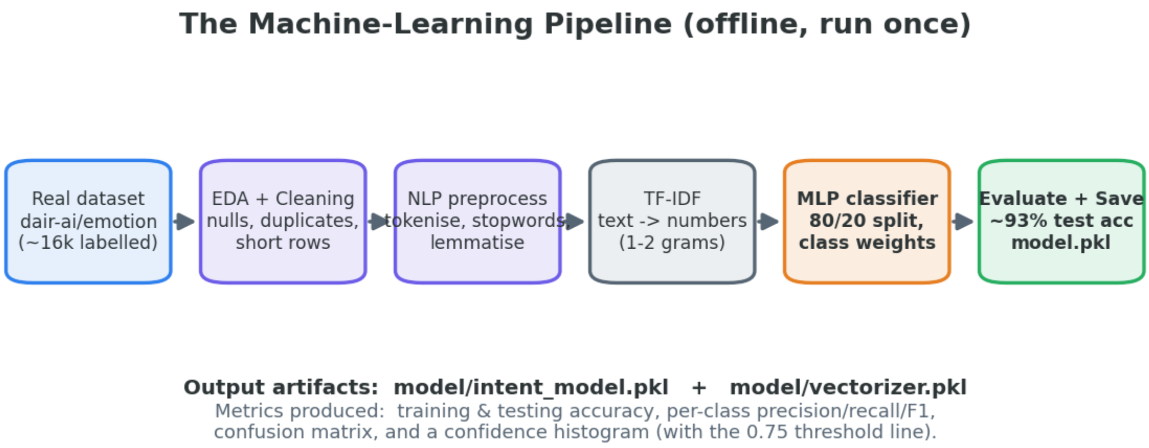


*Figure 2 — Six stages from raw labelled text to a saved, evaluated model.*

## Stage by stage

- **Real dataset** — dair-ai/emotion: ~16,000 short messages, each labelled with an emotion (joy, sadness, anger, fear, love, surprise). We map these onto mental-health intents.
- **EDA + cleaning** — we look at the data first: how many of each class, how long messages are, and remove blanks, duplicates and ultra-short rows that carry no signal.
- **NLP preprocessing** — lower-case, split into tokens, drop common stop-words ('the', 'is'), and lemmatise ('running' → 'run') so similar words count as one.
- **TF-IDF** — converts text to numbers. Each message becomes a vector where words that are frequent *here* but rare *overall* get a high weight. We use 1- and 2-word grams and keep the top 5,000 terms.
- **MLP classifier** — a small neural network (hidden layers 128→64) trained on an 80/20 stratified split, with class weights so rare emotions aren't ignored.
- **Evaluate + save** — about **93% test accuracy**; we save `model/intent_model.pkl` and `model/vectorizer.pkl`.

**Why TF-IDF + MLP instead of a big model like BERT?** Because the proposal is a data-science exercise: TF-IDF + MLP is fast, runs on a laptop with no GPU, and — crucially — is **explainable**. We can point to exact words driving a prediction and show every metric. A huge pre-trained transformer would hide the data-science we're being graded on.

# 4. Runtime decision flow (what happens per message)

When a real message arrives, it passes through a series of checks before any machine-learning guess is trusted. Safety comes first.

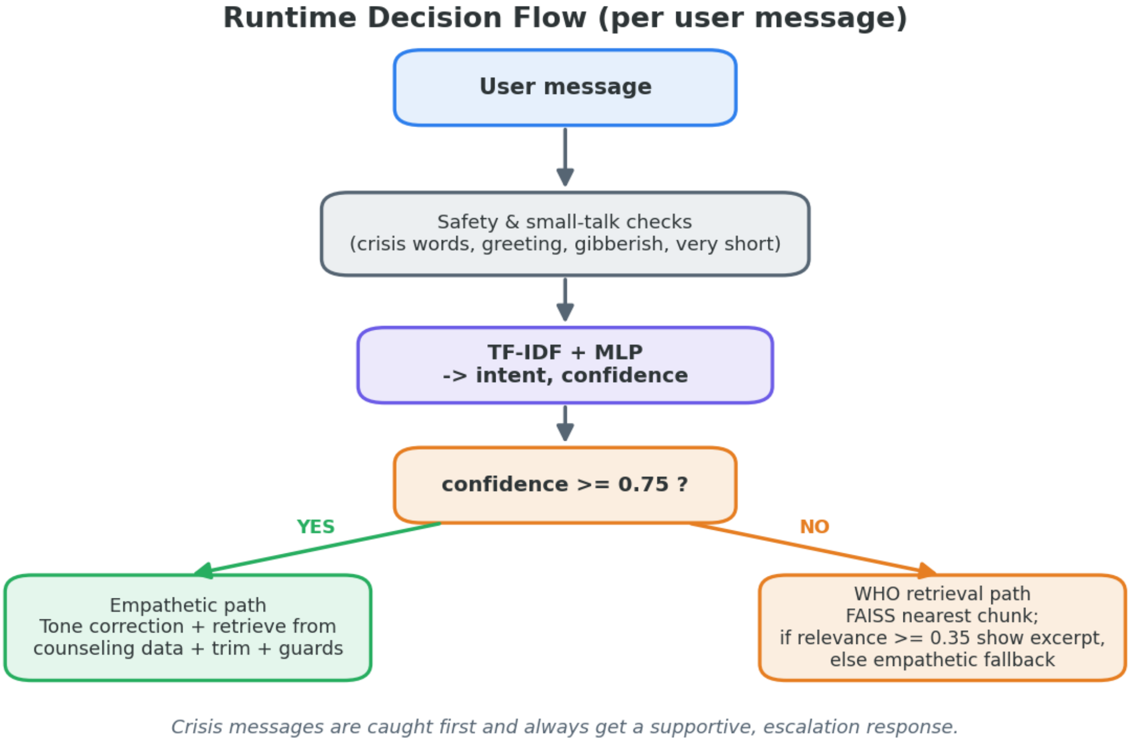


*Figure 3 — Each message is screened for safety and small-talk, classified, then routed by confidence.*

- **Safety & small-talk checks** — crisis words trigger a supportive escalation response immediately. Greetings, gibberish and very short messages get sensible canned replies instead of being force-fed to the model.
- **Classify** — TF-IDF + MLP produce an intent and a confidence (the highest class probability).
- **Confidence** ≥ **0.75** → **empathetic path** — fetch a response from the counseling dataset, correct its tone to match the user, trim it to a couple of sentences, and avoid repeating recent replies.
- **Confidence < 0.75** → **WHO path** — search the guideline index; if the best match scores ≥ 0.35 relevance, show a tidied excerpt; otherwise fall back to empathetic support.

**Why a threshold at all?** A classifier is most useful when it knows when to be humble. If it isn't sure, guessing an emotion and replying confidently is worse than looking up an authoritative answer. 0.75 was chosen from the confidence histogram — it's where correct predictions cluster above and shaky ones below.

# 5. The WHO knowledge base (RAG-Lite)

RAG means **Retrieval-Augmented Generation** — instead of inventing an answer, the system *retrieves* a real passage from trusted documents. Ours is 'RAG-Lite': we retrieve and present authentic WHO text rather than paraphrasing it, so nothing is fabricated.

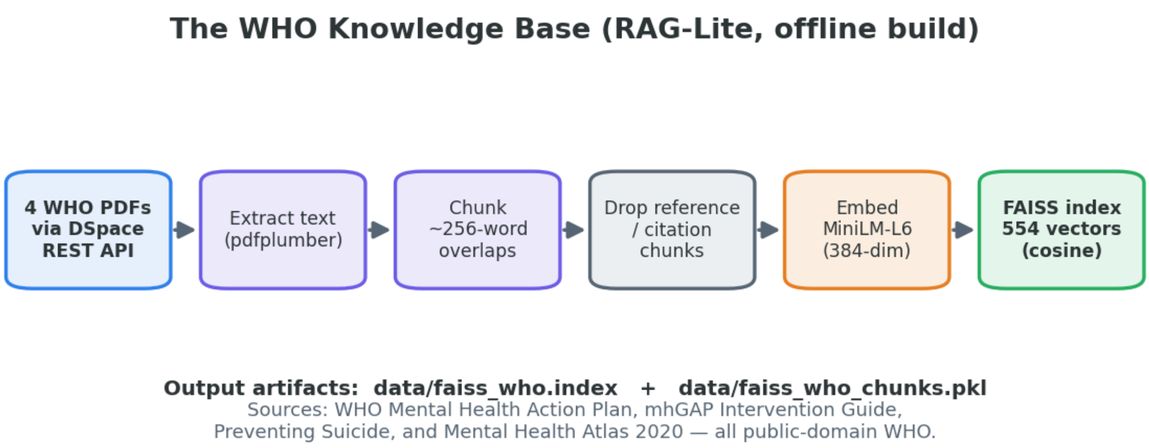


*Figure 4 — Four official WHO PDFs become a searchable index of 554 passages.*

## How the library is built

- **4 WHO PDFs** — downloaded from WHO's official IRIS repository through its DSpace REST API (handles, not fragile direct links).
- **Extract text** — pdfplumber pulls the words out of each PDF.
- **Chunk** — text is split into ~256-word overlapping passages so a search can return a focused snippet.
- **Drop references** — citation/bibliography chunks are detected and removed (151 dropped) so searches don't surface reference lists.
- **Embed** — each chunk is turned into a 384-number vector with the all-MiniLM-L6-v2 sentence-transformer, so meaning (not just words) can be compared.
- **FAISS index** — 554 vectors stored in a FAISS index for instant cosine-similarity search.

**Sources:** WHO Mental Health Action Plan 2013–2030, mhGAP Intervention Guide v2, Preventing Suicide: A Global Imperative, and Mental Health Atlas 2020 — all public WHO documents.

**Relevance gate (0.35):** a search always returns its nearest passage, but 'nearest' can still be irrelevant. If the best score is below 0.35 we discard it and give empathetic support instead — this is what stopped the early bug where unrelated text (e.g. a passage about snakes) was shown for an off-topic message.

# 6. Conversation-quality guardrails

A correct intent isn't enough — the reply has to *feel* right. Several rule-based guards sit on top of the model to keep the conversation natural and safe.

## Tone correction (symmetric)

The emotion model can be fooled by surface words, so we correct it using context. The logic is symmetric and ordered by priority:

- **Negation / negative words win first** — 'I did *not* get the job' → sadness, never joy.
- **Clear positive cues** → **joy** — 'I **won** this competition', '**I got the job**', '**I passed**' → congratulate, don't commiserate.
- **Stress topics** → **fear/stress** — 'I have **exams**', '**deadline** tomorrow' → acknowledge the pressure.
- **Positive sentence but no strong cue** → **neutral** — avoids over-claiming an emotion.

This directly fixes two reported bugs: 'I got final exams' was being read as joy, and 'I won this competition' would have been called stressful. Now tone is judged from the whole situation, both ways.

## Anti-repetition & meta-complaints

- Every reply remembers the **last 6** bot messages and won't repeat them.
- If the user complains ('you keep repeating', 'going in circles') we detect it, apologise, and deliberately change direction.
- After several upbeat turns the bot de-escalates instead of robotically asking the same probing question.

## Length & relevance trimming

- Long forum-style answers are trimmed to ~2 sentences / 55 words.
- Filler openers ('Thank you for reaching out!') and self-references ('As an American…') are stripped — this fixed the 'where did America come from' bug.
- Messages under 6 words get a short, intent-appropriate reply that asks for a little more detail, instead of a wall of text.

# 7. File-by-file map

Where each piece of the project lives:

| Path | What it does |
|---|---|
| `app.py` | The Streamlit web app: UI, session state, safety checks, classification, confidence routing, guardrails, and the two response paths. |
| `model/train_intent.py` | Offline training: loads + cleans the emotion dataset, builds TF-IDF, trains the MLP, evaluates, and saves the .pkl files. |
| `model/humanize.py` | Tone-correction logic (symmetric emotion fixing, stress + positive indicators). |
| `model/faiss_rag.py` | Loads the FAISS index and retrieves the most relevant WHO passage with the 0.35 relevance gate. |
| `model/intent_model.pkl` | Saved trained classifier (committed so the app runs after cloning). |
| `model/vectorizer.pkl` | Saved TF-IDF vectoriser. |
| `data/build_who_corpus.py` | Downloads the 4 WHO PDFs via the DSpace REST API, extracts + chunks text, drops references, embeds, and builds the FAISS index. |
| `data/faiss_who.index` | The 554-vector FAISS similarity index. |
| `data/faiss_who_chunks.pkl` | The text of each indexed passage. |
| `data/who_corpus/*.pdf` | The 4 source WHO PDFs. |
| `docs/diagrams.py` | Generates the four explanatory diagrams. |
| `docs/build_docs.py` | Builds this PDF. |

## 8. Key results

- **~93% test accuracy** on the held-out 20% — the model generalises, not just memorises.
- **Per-class precision / recall / F1** reported for every emotion, so weak classes are visible.
- **Confusion matrix** shows which emotions are mixed up (e.g. fear vs sadness) — expected and explainable.
- **Confidence histogram** with the 0.75 line justifies the routing threshold visually.
- **554 clean WHO passages** indexed (from 705 raw, 151 reference chunks removed).

## 9. Glossary (say these clearly in the viva)

| Term | Plain meaning |
|---|---|
| **Intent** | The underlying meaning/need behind a message (e.g. anxiety, low mood, a factual question). |
| **TF-IDF** | Term Frequency–Inverse Document Frequency: turns text into numbers, weighting words that are frequent in one message but rare across all messages. |
| **MLP** | Multi-Layer Perceptron: a small feed-forward neural network used here as the classifier. |
| **Embedding** | A list of numbers (a vector) that captures the meaning of a piece of text, so similar meanings sit close together. |
| **FAISS** | Facebook AI Similarity Search: a library for finding the nearest vectors very fast. |
| **Cosine similarity** | Measures how close two vectors point in direction — used to rank passages by relevance. |
| **RAG** | Retrieval-Augmented Generation: ground answers in retrieved real documents instead of inventing them. |
| **Confidence threshold** | The 0.75 cutoff that decides whether to trust the classifier or look something up. |
| **Relevance gate** | The 0.35 cutoff a retrieved WHO passage must clear to be shown. |

# 10. Viva question bank (with answers)

**Q. Why TF-IDF + MLP instead of BERT?**

A. It's a data-science project graded on the pipeline, not raw accuracy. TF-IDF + MLP is fast, laptop-friendly (no GPU), and explainable — we can show the exact words and every metric. BERT would hide that work.

**Q. How do you handle class imbalance?**

A. A stratified 80/20 split keeps class proportions in both sets, and we pass class weights (via sample_weight) so rare emotions still influence training.

**Q. What does the confidence score mean?**

A. It's the highest class probability from the MLP's predict_proba. High means the model strongly favours one intent; low means the classes are close and it's unsure.

**Q. Why route on confidence?**

A. A humble classifier is more trustworthy. When unsure, retrieving an authoritative WHO passage beats confidently guessing an emotion.

**Q. How did you pick 0.75 and 0.35?**

A. 0.75 from the confidence histogram — correct predictions cluster above it. 0.35 is the empirical relevance floor below which retrieved passages were off-topic.

**Q. Isn't retrieval just keyword search?**

A. No — we embed text into 384-dim vectors with a sentence-transformer and compare by cosine similarity, so it matches by *meaning*, not exact words.

**Q. How do you avoid fabricated medical advice?**

A. The WHO path shows authentic excerpts verbatim (RAG-Lite), guarded by the 0.35 relevance gate; crisis messages are caught first and always get a safe escalation reply.

**Q. How do you stop repetitive or off-tone replies?**

A. Last-6 memory prevents repeats, a meta-complaint detector changes direction when the user notices, symmetric tone-correction reads the whole situation, and trimming keeps replies short and on-topic.

**Q. What are the limitations?**

A. It's not a medical tool; the emotion dataset is general (not clinical); TF-IDF ignores word order beyond bigrams; and the knowledge base is only four documents.

**Q. Could you improve it?**

A. Add more WHO/clinical sources, try calibrated probabilities, expand to multi-turn context, and A/B test the thresholds with user feedback.

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*End of notes — MindSpace DS. Built with real data, standard techniques, and explainable choices.*